

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT (3 Questions)

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO5 – Precision (BL6)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	30 (3 Questions × 10 Marks)
CO5 Statement:	<i>Create effective and interactive visualizations by integrating data wrangling and advanced visualization tools (BL6)</i>		
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Section / Batch:	AI&DS	Date:	

Code of Conduct

I G Kushwanth(192424150) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: G Kushwanth

Instructions

- This test contains 3 questions, each carrying 10 marks. Total: 30 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus Area	Q. Nos.	Marks
Visualizing Uncertainty	Probability visualization, confidence intervals, uncertainty interpretation	1	1
Handling Overplotting	Overplotting, transparency, jittering, alternative plots	2	1
Principles of Effective Visualization & Accessibility	Proportional ink principle, misleading axes, accessible color design	3	1
GRAND TOTAL:		30 Marks	

Annexure A – Scenario Based Assessment

1. A weather forecasting agency predicts rainfall probability (%) for a week along with uncertainty:

Day	Rainfall Probability	Confidence Interval
Mon	60	10
Tue	65	20
Wed	70	5
Thu	55	15
Fri	80	25

As a data analyst, design a visualization that clearly communicates both probability and uncertainty. Which day's prediction is most reliable? Justify your answer using visualization principles.

Answer:

Visualization Design: Use a bar/point chart showing rainfall probability for each day, with error bars (confidence intervals) overlaid on each point. The error bars extend above and below the probability value by the size of the CI, and a lighter shaded band can also be drawn to show the possible range visually. This way, both the estimate (probability) and its uncertainty (CI) are communicated in a single chart, instead of showing only the bare probability values as a plain bar chart would.

Most Reliable Day: Wednesday is the most reliable prediction. It has a rainfall probability of 70% with the narrowest confidence interval of ± 5 , meaning the true value is very likely to lie between 65% and 75%. In contrast, Friday shows the highest probability (80%) but the widest interval (± 25), so its true value could range from 55% to 100%, making it far less certain despite the higher headline number.

Justification: In uncertainty visualization, reliability depends on the precision of a value, not just its magnitude. A moderate probability with a tight confidence interval (Wednesday) reflects a stable, well-supported model estimate, while a high probability with a wide interval (Friday) indicates the model itself is unsure. Displaying confidence intervals alongside probabilities (rather than probabilities alone) prevents viewers from over-trusting forecasts that are numerically high but statistically uncertain, which is the core goal of uncertainty visualization.

2. A mobile app company collects user ratings (1–5) from 500 users. Many users give the same ratings:

Rating	Frequency
1	50
2	80
3	150
4	140
5	80

If plotted directly, the visualization suffers from overplotting. Propose a suitable visualization technique and explain how transparency or jittering can improve clarity. Suggest an alternative plot if needed.

Answer:

Problem: Since ratings only take 5 discrete values, plotting all 500 individual data points directly (e.g., as a scatter or strip plot) causes heavy overplotting — hundreds of points stack exactly on top of one another at each of the 5 x-positions, so it becomes impossible to visually judge how many users chose each rating.

Recommended Technique: The most suitable visualization is a bar chart of counts (a frequency plot) for each rating value 1–5. Because the data is aggregated into counts before plotting, no individual points overlap, and bar height/length directly and clearly encodes how many users gave each rating.

Role of Transparency and Jittering (for point-based plots): If individual points must still be shown, transparency (alpha blending, e.g., 0.1–0.3 opacity) makes overlapping points visually darker where density is high, giving a rough sense of frequency. Jittering adds a small amount of random noise to each point's position so points that would otherwise sit on the exact same spot are spread out slightly, letting the eye distinguish clusters instead of one solid mass of overlapping dots.

Alternative Plot: If deeper distributional insight is needed (e.g., comparing rating distributions across different user segments or app versions), a violin plot or box plot can be used. These summarize the shape, spread, and central tendency of the ratings compactly, avoiding overplotting entirely while conveying more nuance than a single bar chart.

3. A company presents employee satisfaction scores:

Department	Score (%)
HR	70
Sales	75
IT	78
Finance	80

The chart uses a truncated y-axis (starting at 65) and red-green color coding. Identify the issues in this visualization. Redesign it using proper axis design, proportional ink principle, and accessible color choices. Explain how your design avoids misleading interpretation.

Answer:

Issues Identified:

- 1. Truncated Y-axis (starting at 65%):** This exaggerates the visual gap between departments — a department at 70% looks much shorter than one at 80%, even though the real difference is only 10 percentage points. This breaks the proportional ink principle, which requires that the visual area/length used to represent a value be proportional to the value itself.
- 2. Red-green color coding:** This is inaccessible to viewers with red-green color blindness (around 8% of men), who cannot reliably distinguish which department the color is signaling as "good" or "bad".

Redesign:

Axis Design: Start the y-axis at 0% so bar heights truly reflect the underlying scores, with no department's performance visually exaggerated or minimized.

Proportional Ink Principle: Use plain, flat 2D bars (not 3D or heavily shaded) whose length scales linearly with the score, and add data labels showing the exact percentage on each bar for precision.

Accessible Color Choices: Replace the red-green scheme with a single consistent, colorblind-safe color (e.g., blue) for all bars, or use a colorblind-friendly diverging palette such as blue-to-orange if a comparison encoding is still needed.

How This Avoids Misleading Interpretation: Starting the axis at zero keeps bar area proportional to the true score, so viewers perceive the actual magnitude of difference between departments rather than an exaggerated one. Using an accessible, non-judgmental color scheme (or removing color-based meaning altogether) ensures the chart communicates the same, correct information to every viewer regardless of color vision, without implying value judgments that could bias interpretation.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks	
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding ; misses key aspects	Misinterprets or fails to understand the scenario		
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues		
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts		
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided		
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____